COMP-421 Database System
School of Computer Science, McGill University
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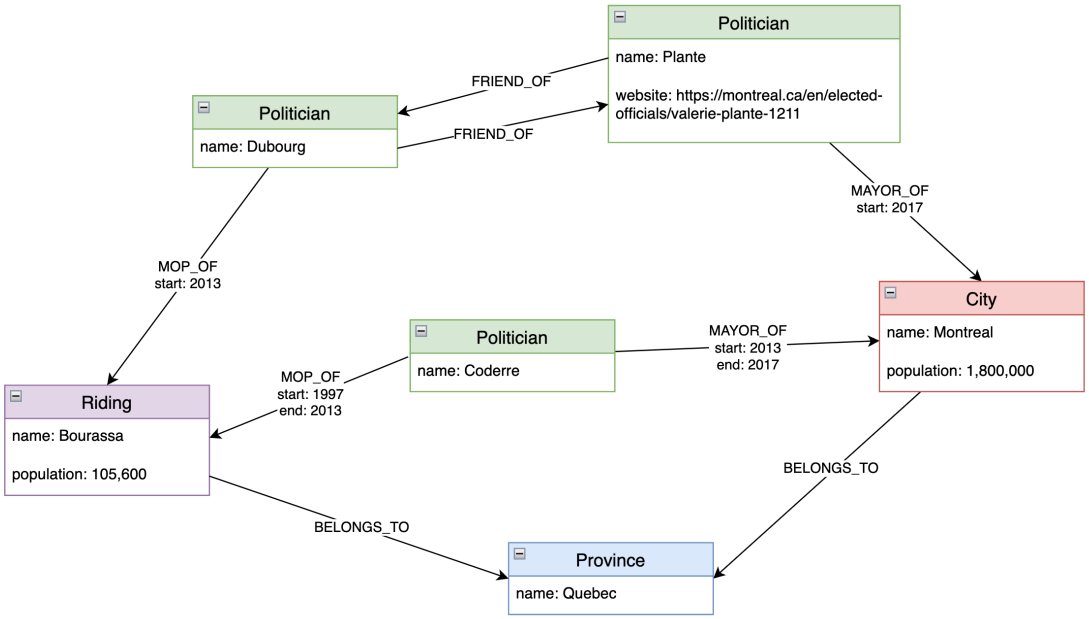
Jocelyne Li (261178604)

Assignment 3: Graph Databases and Query Evaluation

1 Part I: Graph Databases

1.1 Ex. 1 —Modeling Graph Databases vs. E/R (15 points)

Graph database:



Cypher query to find politicians who have been mayor of Montreal twice, with at least one other person serving as mayor in between:

```
MATCH (p:Politician)-[m1:MAYOR_OF]->(city:City {name: "Montreal"}),
      (p)-[m2:MAYOR_OF]->(city),
      (q:Politician)-[m_between:MAYOR_OF]->(city)
WHERE m1 <> m2
  AND m1.end < m_between.start AND m_between.end < m2.start
  AND p <> q
RETURN DISTINCT p.name AS PoliticianWhoWasMayorTwice;
```

Entity Sets and Primary Keys

- **Politician**(pid, name, website)
  name is not unique, so pid is used as surrogate key
- **City**(cid, name, population)
  name is not unique, so cid is primary key

- **Riding**(name, population)
  name is unique, used as primary key
- **Province**(name)
  name is unique

Relationship Sets and Keys

- **MayorOf**(pid, cid, start, end)
  Between Politician and City, with start and end year
- **MoPOf**(pid, riding_name, start, end)
  Between Politician and Riding, with start and end year
- **BelongsTo_City**(cid, province_name)
  Each city belongs to one province
- **BelongsTo_Riding**(riding_name, province_name)
  Each riding belongs to one province
- **Friendship**(pid1, pid2)
  Symmetric relationship between two politicians

1.2 Ex. 2 —Cypher Queries (15 points)

1.(0 points) Find all 300-level courses.

```
MATCH (c:Course)-[:IS_OF_TYPE]->(t:Type {name: "300-level"})
RETURN DISTINCT c.name, c.title;
```

2.(0 points) Find all pairs of students that are either friends or roommates.

```
MATCH (s1:Student) -[r]->(s2:Student)
WHERE type(r) IN["FRIEND_OF", "ROOMMATES"]
RETURN DISTINCT s1.name AS Student1, s2.name AS Student2;
```

3.(0 points) Find all pairs of students that are roommates but not friends

```
MATCH (s1:Student)-[:ROOMMATES]->(s2:Student)
WHERE NOT (s1)-[:FRIEND_OF]->(s2) AND NOT (s2)-[:FRIEND_OF]->(s1)
RETURN DISTINCT s1.name AS Student1, s2.name AS Student2;
```

4.(5 points) Find all pairs of students that are friends and roommates.

```
MATCH (s1:Student)-[:FRIEND_OF]->(s2:Student),
      (s1)-[:ROOMMATES]->(s2)
RETURN DISTINCT s1.name AS Student1, s2.name AS Student2;
```

5.(5 Points) Find the names of the students who have taken courses at a different university than the one they study at.

```
MATCH (s:Student)-[:TAKES]->(c:Course)-[:GIVEN_AT]->(cu:University),
      (s)-[:STUDY_AT]->(su:University)
WHERE cu.name <> su.name
RETURN DISTINCT s.name;
```

6.(5 Points) Find pairs of students that take the same course and have a linkage through friend relationships (i.e., there is a path of edges labelled FRIEND OF that connects those two students).

```
MATCH (s1:Student)-[:TAKES]->(c:Course)<-[:TAKES]-(s2:Student)
WHERE s1 <> s2 AND EXISTS {
  MATCH path = (s1)-[:FRIEND_OF*]-(s2)
  RETURN path
}
RETURN DISTINCT s1.name AS Student1, s2.name AS Student2, c.name AS
↪  SharedCourse;
```

2 Part II: Query Evaluation

2.1 Ex. 3 —Index Sizes (10 points)

1.total number of data entries, the number of rids per data entry, and the size of a data entry

- **Total number of data entries:** Since `amount` has 3,000 distinct values, we have:

$$\text{Data entries} = 3,000$$

- **Number of RIDs per data entry:** There are 120,000 payments, distributed uniformly over 3,000 amount values:

$$\frac{120,000 \text{ payments}}{3,000 \text{ values}} = 40 \text{ RIDs per entry}$$

- **Size of a data entry:**
  - Key (amount): `INT` = 4 bytes
  - Each RID = 10 bytes
  - Total size:

$$4 + (40 \times 10) = \boxed{404 \text{ bytes}}$$

2.the number of leaf pages and the height of the tree

- **Usable space per page:**

$$4,000 \times 0.75 = 3,000 \text{ bytes}$$

- **Data entries per page:**

$$\left\lfloor \frac{3,000}{404} \right\rfloor = 7$$

- **Leaf pages:**

$$\left\lceil \frac{3,000}{7} \right\rceil = \boxed{429}$$

- **Internal node entry size:**
  - Key (INT) = 4 bytes, pointer = 6 bytes → entry = 10 bytes
  - Entries per internal node:
$$\left\lfloor \frac{3,000}{10} \right\rfloor = 300$$
- **Internal pages:**
$$\left\lceil \frac{429 \text{ leaf pages}}{300} \right\rceil = 2$$
- **Root page:**
$$1 \text{ (always fits)}$$
- **Total tree height:**
$$\text{Root (1)} \rightarrow \text{Internal (2)} \rightarrow \text{Leaves (429)} \Rightarrow \boxed{3}$$

2.2 Ex. 4 —Using indices (25 points)

```
SELECT *
FROM Payment
WHERE paydate = '2025-03-05' AND amount > 2750;
```

1.Indicate the access costs (in number of pages retrieved leading to I/O) for this query in each of the scenarios for the case that D=2025-03-05 and M=2750. Provide a simple calculation that shows how you have derived the values.

Assumptions and setup

- Total `Payment` records: 120,000
- Records are evenly distributed across 365 days:
$$\frac{120{,}000}{365} \approx 329 \text{ records on 2025-03-05}$$
- `amount` has 3,000 distinct values, uniformly distributed
- Fraction of payments with `amount` > 2750:
$$\frac{250}{3000} = \frac{1}{12}$$
- Estimated number of qualifying records:
$$329 \times \frac{1}{12} \approx 27.4 \approx 28 \text{ records}$$
- `Payment` table is stored in 1,000 pages

(a)(4 Points) you do not use any index.

Full scan of the `Payment` table.

$$\boxed{1000 \text{ I/Os}}$$

(b)(6 Points) you use the index on paydate (see above).

- We are using a Type II B^+-tree index on `paydate`, which is **unclustered**.
- There are 365 distinct `paydate` values stored in the index.
- The date `'2025-03-05'` corresponds to one leaf page, which points to **329 RIDs**.

Approach 1: Worst-case analysis

- Each RID could point to a different data page (no overlap).
- So, we must fetch all 329 pages plus 1 leaf page from the index.

$$\text{Cost} = 1 \text{ (index leaf)} + 329 \text{ (data pages)} = \boxed{330 \text{ I/Os}}$$

Approach 2: Smarter estimate

- Assume only $\frac{1}{12}$ of the 329 records satisfy the second condition `amount > 2750`.
- That gives approximately 28 matching records.
- Worst case, these 28 records are on 28 different data pages.

$$\text{Cost} = 1 \text{ (index leaf)} + 28 \text{ (data pages)} = \boxed{29 \text{ I/Os}}$$

(c)(6 Points) you use the index on amount (see above).

- Values $> 2750 \Rightarrow 250$ distinct `amount` values
- Each value: 40 RIDs $\Rightarrow 250 \times 40 = 10{,}000$ RIDs
- All must be checked for `paydate` = '2025-03-05'
- Each RID could point to a different page

$$250 \text{ (leaf pages)} + 10{,}000 \text{ (data pages)} = \boxed{10{,}250 \text{ I/Os}}$$

(d)(4 Points) you use both indexes.

- Intersect RIDs from both indexes in memory
- Same match count (≈ 28 records)
- Cost: 1 (`paydate` leaf) + 250 (`amount` leaves) + 28 data pages

$$1 + 250 + 28 = \boxed{279 \text{ I/Os}}$$

2.(0 Points) Calculate this also for general values of D and M.

Let:

- R = total records in `Payment` (120,000)
- V = distinct `amount` values (3,000)
- D = chosen date
- M = chosen threshold amount

Then:

$$\text{Records on } D = \frac{R}{365}, \quad \text{Fraction satisfying } \texttt{amount > M} = \frac{V - M}{V}$$

$$\Rightarrow \text{Expected result size} = \frac{R}{365} \times \frac{V - M}{V}$$

3.(5 Points) What would be the access costs if the index on paydate was clustered? Give a short calculation that explains your answer.

In a clustered index, the records with the same `paydate` are stored physically close together in sorted order on disk. This means:

- There are approximately 329 payments made on `2025-03-05`.
- Each payment record is estimated to be around 30 bytes in size.
- With a 75% page fill factor, each 4,000-byte page holds:

$$\left\lfloor \frac{3000}{30} \right\rfloor = 100 \text{ records per page}$$

- Therefore, 329 records span approximately:

$$\left\lceil \frac{329}{100} \right\rceil = 4 \text{ contiguous pages}$$

- Since we are only interested in those with `amount > 2750` (roughly 1/12 of them), only ≈ 28 records will match.
- These 28 qualifying records are likely concentrated within only **2 to 3 of those 4 pages** (not spread across all of them), due to physical locality from clustering.

Hence, the estimated cost is:

$$\text{Cost} \approx 1 \text{ (index leaf)} + 2\text{–}3 \text{ data pages} = \boxed{3\text{–}4 \text{ I/Os}}$$

2.3 Ex. 5 —Joins (15 Points)

Assume now a join between Account and Payment (WHERE Account.accountid = Payment.srcaccountid).

Assume an unclustered type II index on accountid of Account.

Indicate

Given:

- `Account`: 8,000 tuples ⇒ 250 pages
- `Payment`: 120,000 tuples ⇒ 1,000 pages
- Each `Payment.srcaccountid` matches 1 `Account`
- Unclustered index on `Account.accountid`: lookup cost = 2 I/Os (1 index page + 1 data page)

1.(0 points) the number of output tuples.

Each payment has one `srcaccountid`, and all are valid references:

$$\boxed{\text{120,000 output tuples}}$$

2.(5 Points) the estimated number of I/O when using an index nested loop join.

- Outer = `Payment` (1000 pages, 120,000 tuples)
- Inner = `Account` (indexed on `accountid`, type II unclustered)

The index must be on the inner relation, so we use `Account` as the inner relation.

- For each of the 120,000 tuples in `Payment`, we perform:
  - 1 I/O to access the index leaf page
  - $\frac{1}{15}$ I/O to fetch the matching data page in `Account` (since 15 records fit per page)
- Total cost per tuple: $1 + \frac{1}{15} = \frac{16}{15}$

$$\text{Total I/Os} = 1000 \text{ (to scan Payment)} + 120{,}000 \times \frac{16}{15} = 1000 + 128{,}000 = \boxed{\text{129,000 I/Os}}$$

3.(5 Points) the estimated number of I/O when using block nested loop join and Account is the outer relation.

- Outer = `Account` (250 pages), Inner = `Payment` (1000 pages), $B = 52$

$$\left\lceil \frac{250}{B-2} \right\rceil = \left\lceil \frac{250}{50} \right\rceil = 5 \text{ iterations}$$

$$\text{Cost} = 250 + 5 \times 1000 = \boxed{\text{5,250 I/Os}}$$

4.(5 Points) the estimated number of I/O when using sort merge join. Assume relations are not already sorted on the join attribute. Hint:- Remember we used a technique in class to reduce the cost of merge-join so that it is less than the sum of the individual costs of merge sort and join.

Assume neither `Account` nor `Payment` is sorted on the join attribute. We use the standard sort-merge join with pipelining, which allows us to combine the final sort pass with the merge join to reduce I/O cost.

Step-by-step cost:

- **Pass 0:** Read and write each relation once to produce sorted runs.

$$2 \times \texttt{AccountPages} + 2 \times \texttt{PaymentPages} = 2 \times 250 + 2 \times 1000 = 2500$$

- **Pass 1 and Merge Join (Pipelined):** Read the sorted runs again and perform the merge join simultaneously without writing the sorted output back to disk.

$$1 \times \texttt{AccountPages} + 1 \times \texttt{PaymentPages} = 250 + 1000 = 1250$$

- **Total I/O Cost:**

$$2500 + 1250 = \boxed{3750 \text{ I/Os}}$$

Note: The optimization here avoids an extra read/write step after sorting, by performing the merge join as we read the sorted data.